

FINAL PRESENTATION

2 0 2 6 A H M I

CHANGING
LANES

L 3 a u t o m a t i o n

Index

01	Research
02	Ideation
03	Implement
04	Prototype
05	Testing Activity

Index

01

Research

Understanding user trust, expectations, and concerns about automation.

02

Ideation

Translating research insights into trust-centered interaction concepts.

03

Implement

Developing interface flows and adaptive interaction strategies.

04

Prototype

Evaluating the proposed HMI through interactive prototypes and user feedback.

05

Testing Activity

Evaluation methods used to validate and refine the concept.

Research

Interview

Selection of People

Quindi



42 yrs

Bea



27 yrs

Eva



57 yrs

Sara



23 yrs

Yijun



55 yrs

Tang



57 yrs

Kaushal



24 yrs

Jaiden



22 yrs

Marta



24 yrs

Paloma



20 yrs

Answers 1/2

“I don’t trust autonomous vehicles because they cannot react fast if something external changes.”

“Trust depends on experience and proven testing.”

NAME	AGE	COUNTRY	OPINION ABOUT AUTONOMOUS VEHICLES	KEY TAKEAWAY
Bea	27	México	She doesn't trust the readiness of the technology for her context where unpredictable behaviours can appear. "My driving context is very rude and the car cannot prevent an accident as a human would do."	Driver context and unpredictability strongly influence trust .
Quindi	42	Colombia	It's best not to replace the driver because they might fall asleep. She thinks autonomous cars prevent people from learning how to drive and react fast. "I don't trust autonomous vehicles because they cannot react fast if something external changes; I think they are useful as systems of support like copilots."	Trust is higher for support systems than full automation; reaction speed is key.
Sara	23	Norway	Views automation mainly as driver support, not full self-driving. Comfortable with assistive features. Trust depends on experience and proven testing. Prefers clear alerts. Open to automation in simple situations, cautious in complex ones.	Users prefer practical, safety-oriented features. Experience and familiarity strongly influence trust . Handover is cognitively demanding.
Eva	57	Norway	Skeptical about autonomous driving and questions its practical value. Believes automation may be safe, but has no strong personal need. Prefers clear safety alerts. Automation useful if it allows doing other tasks while driving.	Users prioritise basic safety over advanced features. Trust depends on proven safety and clear communication. Handover can cause stress.
Yijun	55	China	Autonomous driving likely to become mainstream. Trust depends on safety improvements and reliable brands. Most valuable for reducing driver fatigue, especially on highways.	Safety is the most critical factor. Automation acceptable in predictable scenarios. Clear handover warnings essential. Positive attitudes encourage adoption.

Answers 1/2

“If it is unclear what the system can handle, confidence quickly decreases.”

“I feel uncomfortable not being in control.”

Tang	57	China	Autonomous driving seen as an important technological trend. Assisted driving improves comfort. Trust influenced by technology development and brand reputation. Open to automation in many situations.	Experience with L2 systems increases acceptance. Automation reduces workload and fatigue. Trust in technology and brand is key. Drivers expect gradual upgrades to L3.
Kaushal	24	India	Open to autonomous driving but has limited understanding. Values safety, comfort, and brand. Trust depends on clear safety guarantees. Prefers predictable, gradual handovers.	Limited understanding shows need for clearer education. Trust and reliability guarantees are critical. Clear handover alerts reduce uncertainty. Local driving conditions influence adoption.
Jaiden	22	India	Skeptical about current driver-assistance systems due to unreliable behavior in city traffic. Values safety, power, comfort. Believes fully autonomous vehicles should drive completely on their own. Prefers to monitor surroundings. Interested in cooperative vehicle communication.	Negative experiences reduce trust . Urban conditions limit usefulness. Vehicle-to-vehicle communication could improve trust . Gradual adoption more likely when technology is proven.
Marta	24	Spain	Limited driving experience, basic car without assistance. Understands autonomous vehicles generally. Low trust , prefers manual driving. Sudden takeovers stressful. Would try autonomous driving only in simple, low-traffic environments.	Lack of experience reduces trust . Handover situations are stressful. Automation more acceptable in low-demand scenarios. Direct experience may increase acceptance.
Paloma	20	Spain	Views autonomous vehicles as fully self-driving but believes trust requires reliability and long-term testing. Uncomfortable not being in control. Automation useful in stressful or repetitive driving. Drivers should remain aware and ready to take control.	Trust depends on reliability, safety, brand. Users expect transparent communication. Automation accepted when it reduces stress or fatigue. Allowing driver control increases perceived safety.

Insights & Opportunities

01

Loss of Control Creates Anxiety

"I don't trust autonomous vehicles because they cannot react fast if something external changes."
"I feel uncomfortable not being in control."
"Sudden takeovers are stressful."



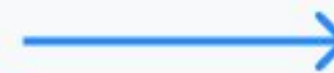
01

Users need to feel continuously involved, not suddenly excluded.

02

Users Prefer Assistance Over Replacement

"Useful as systems of support like copilots."
"Automation should help in stressful situations."
"Assisted driving improves comfort."



02

People trust collaborative automation more than fully independent automation.

03

Trust Is Strongly Linked to Predictability

"Clear handover warnings are essential."
"Automation should help in stressful situations."
"Automation acceptable in predictable scenarios."



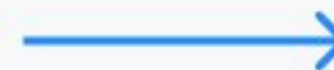
03

The handover experience must feel calm, predictable, and guided.

04

Trust Depends on Understanding System Behaviour

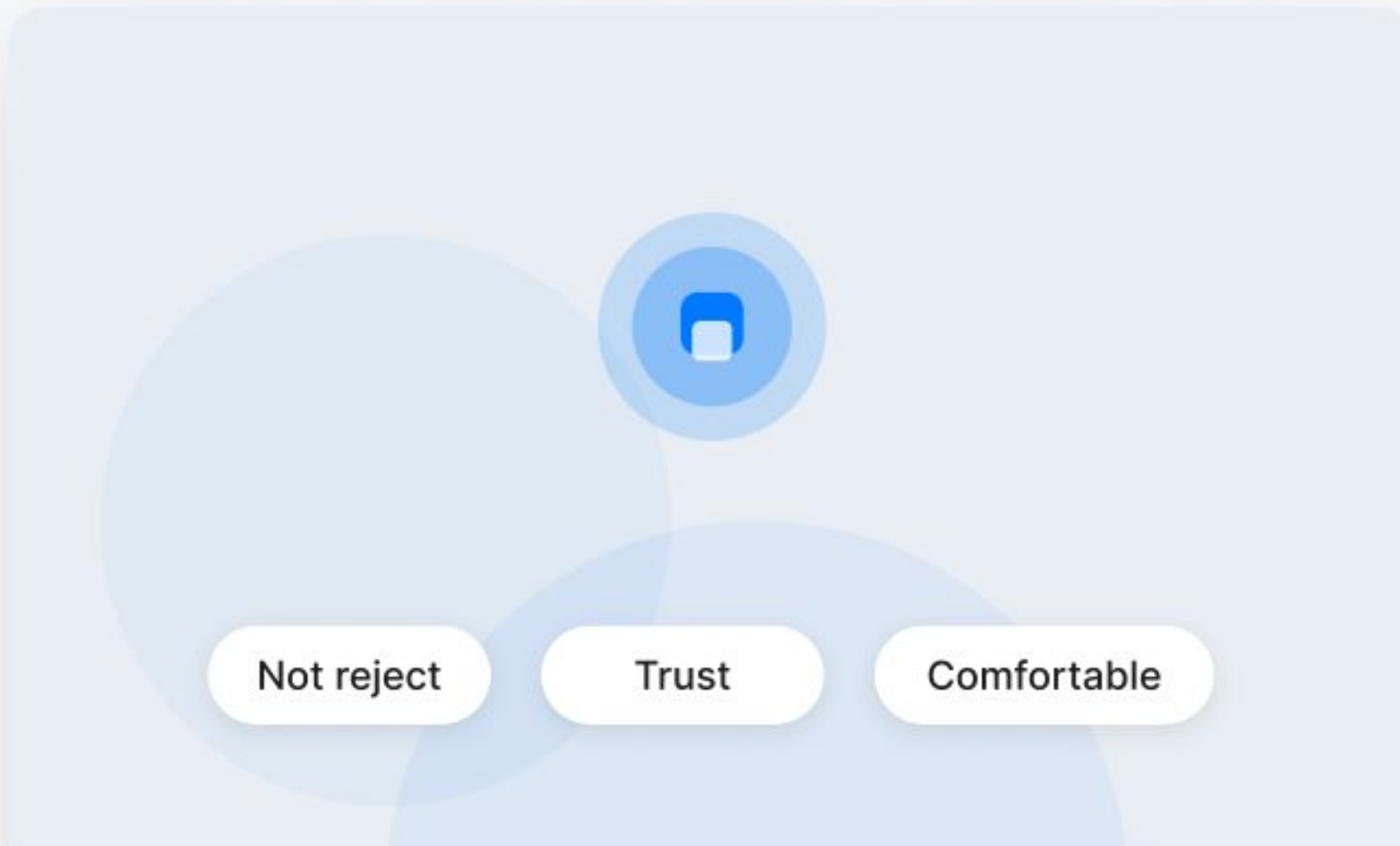
"Trust depends on experience and proven testing."
"If it is unclear what the system can handle, confidence quickly decreases."
"Drivers should remain aware and ready to take control."



04

Transparency builds trust. Users need to understand what the system sees, what it plans to do and why it asks for intervention.

Takeaway



Not reject Trust Comfortable

Observed Patterns

- Drivers are **not rejecting automation itself**. They are rejecting uncertainty, loss of control and unclear system behaviour.
- **Trust** increases when the system communicates clearly, behaves predictably, explains decisions and supports the driver.
- Drivers are most **comfortable** in predictable driving scenarios, when they remain mentally involved and when handovers are gradual.



Identified a user group and focus on middle-aged drivers.

User group Middle-aged drivers

Emerging Opportunity

- Identified a **user group** that still drives actively today, is capable of adapting to new technology, but is beginning to think about long-term driving safety and independence.
- The **middle-aged drivers** because this group sits at a critical transition point. Experienced enough to recognise driving limitations, young enough to still adapt to new systems, and motivated by maintaining independence later in life.

Summary

This revealed that trust is not built in a single interaction. It develops gradually over time through understanding, predictability, and support.

Literature Review

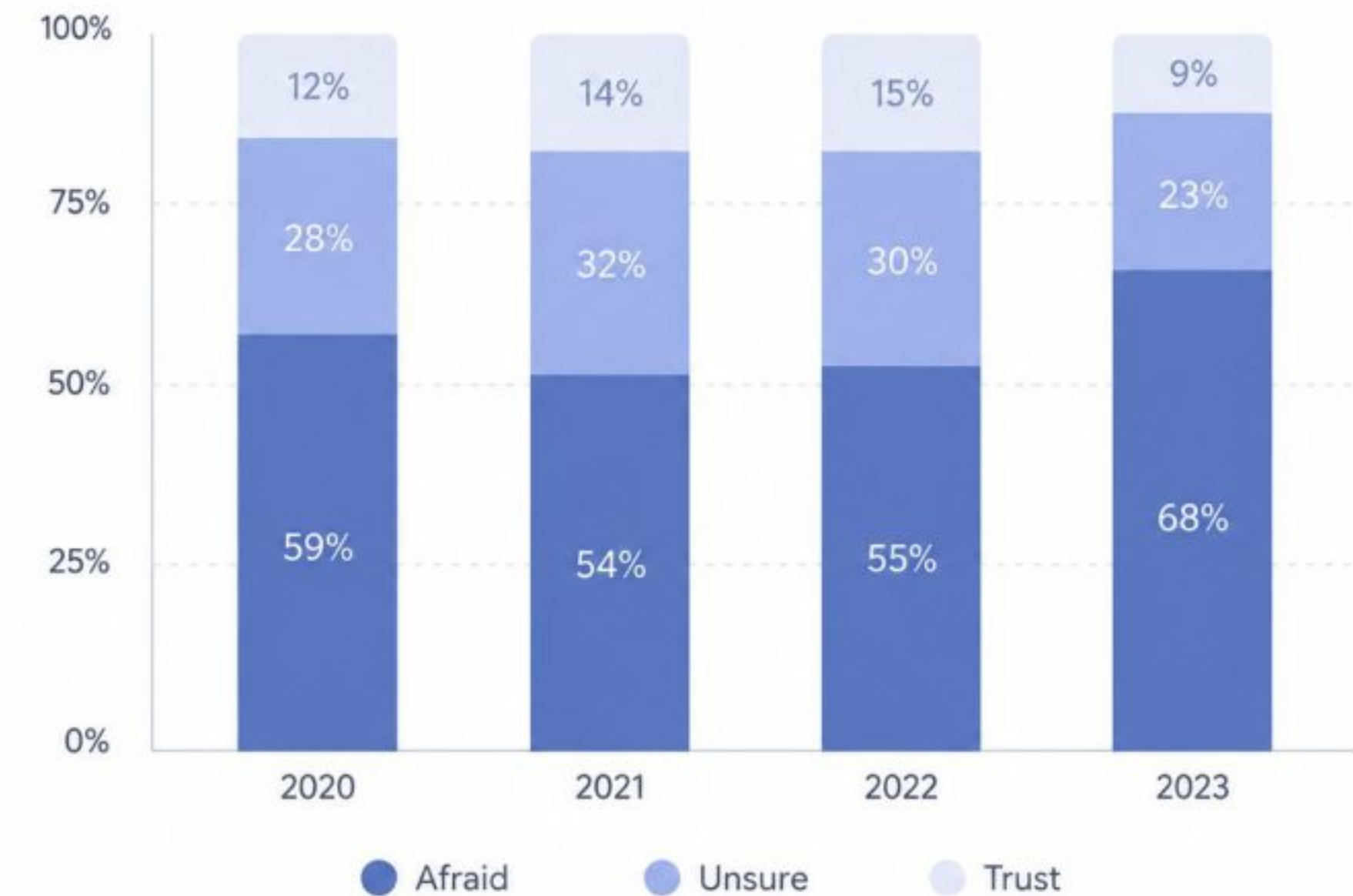
Results of AAA's (the American Automobile Association) annual automated vehicle survey

“These findings suggest **improvements are still needed to build public trust and knowledge surrounding emerging vehicle technology.**

There is also a need to **dispel confusion around automated vehicles.** AAA's survey found that nearly one in ten drivers believe they can buy a vehicle that drives itself while they sleep.”

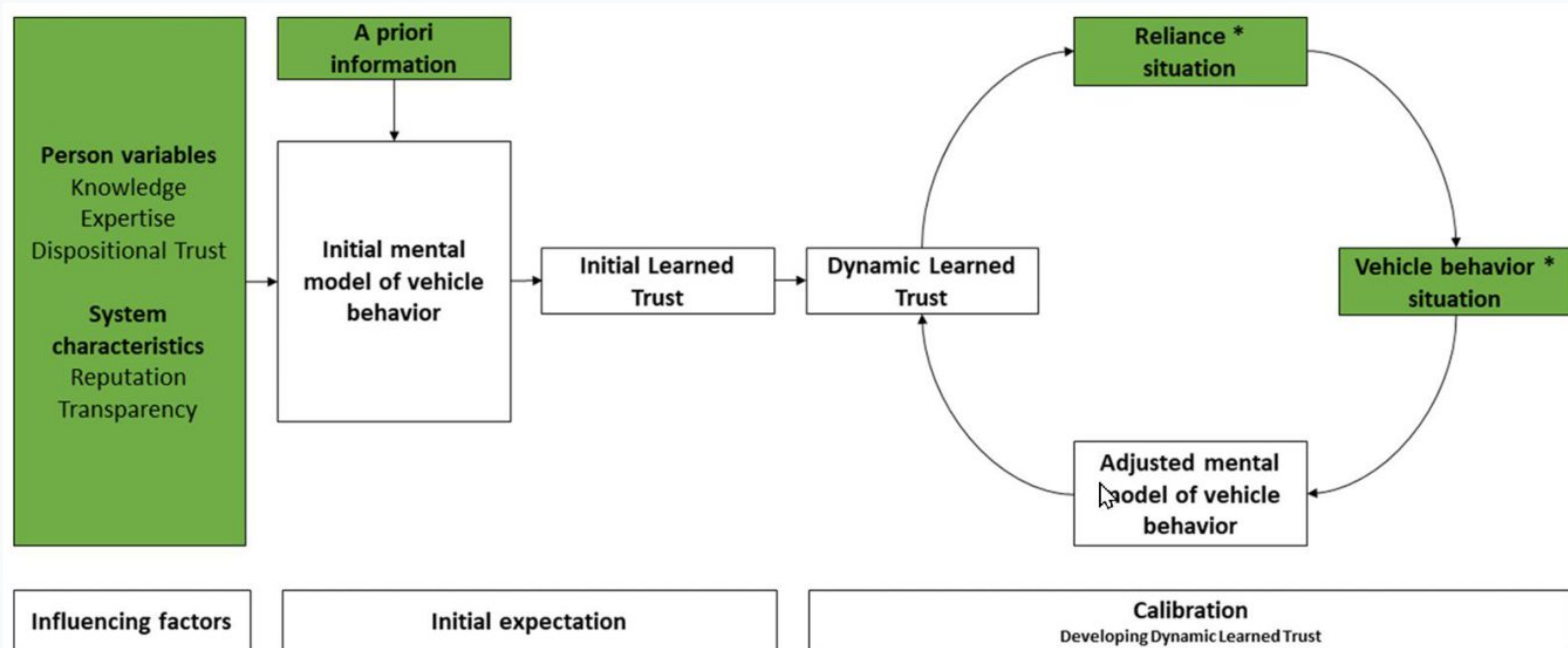
—Taken from: AAA: Fear of Self-Driving Cars on the Rise

Driver Attitudes Toward Self-Driving Vehicles



Conceptual framework of the development of trust toward a specific automated driving system

→ Trust calibration



—Taken from: F. Walker et al., "Trust in automated vehicles: constructs, psychological processes, and assessment," *Front. Psychol.*, vol. 14, Nov. 2023, doi: 10.3389/fpsyg.2023.1279271.

Ideation

User Persona

Meet Ingrid



“ I want to stay independent and feel safe driving for as long as possible ”



Age

52



Country

Norway



Occupation

Senior HR Advisor (Public Sector)



Residence

Suburban area outside Oslo



Family

Married, two adult children who recently moved out



Driving Profile

Daily commuter, mostly highway and suburban driving

A Shift in Perspective

After watching her 75-year-old mother become less confident in traffic, Ingrid began reflecting on her own future as a driver. For the first time, **Level 3 automation feels relevant**. Not as a replacement for driving, but as a way to **maintain freedom and independence longer**.

What Creates Uncertainty

For Ingrid, the biggest barrier is not the technology itself, but the feeling of **losing control**. Sudden handovers, unclear system behaviour, and complex interfaces quickly reduce her trust in automation.

What She Needs

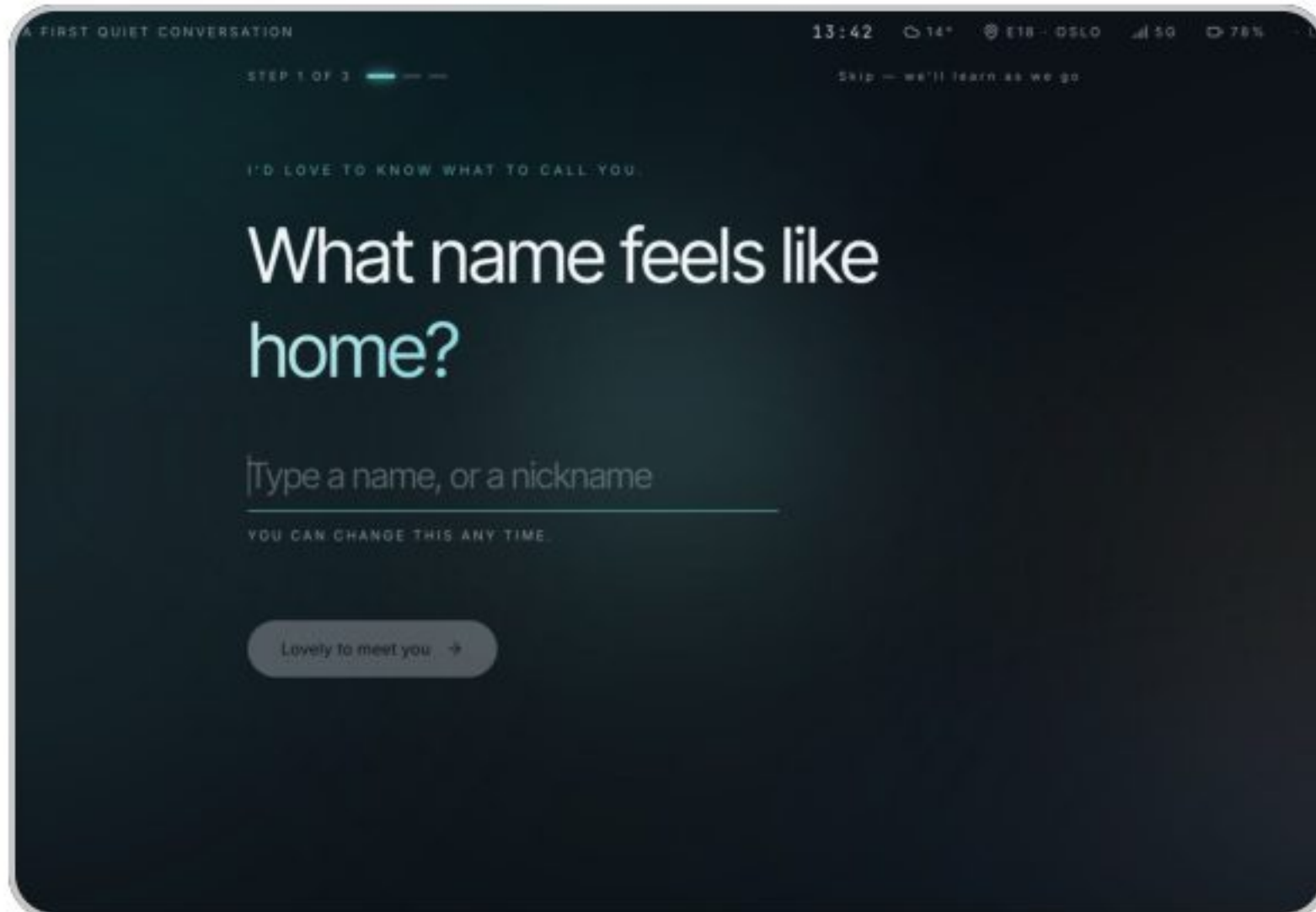
To build trust, Ingrid needs a system that feels **clear, predictable, and supportive**. The transition between automated and manual driving should happen gradually, giving her time to regain **situational awareness and confidence**.

How Might We

How might we design Level 3 interactions that **build trust over time**, so drivers feel confident relying on the system later in life?

Concept Description

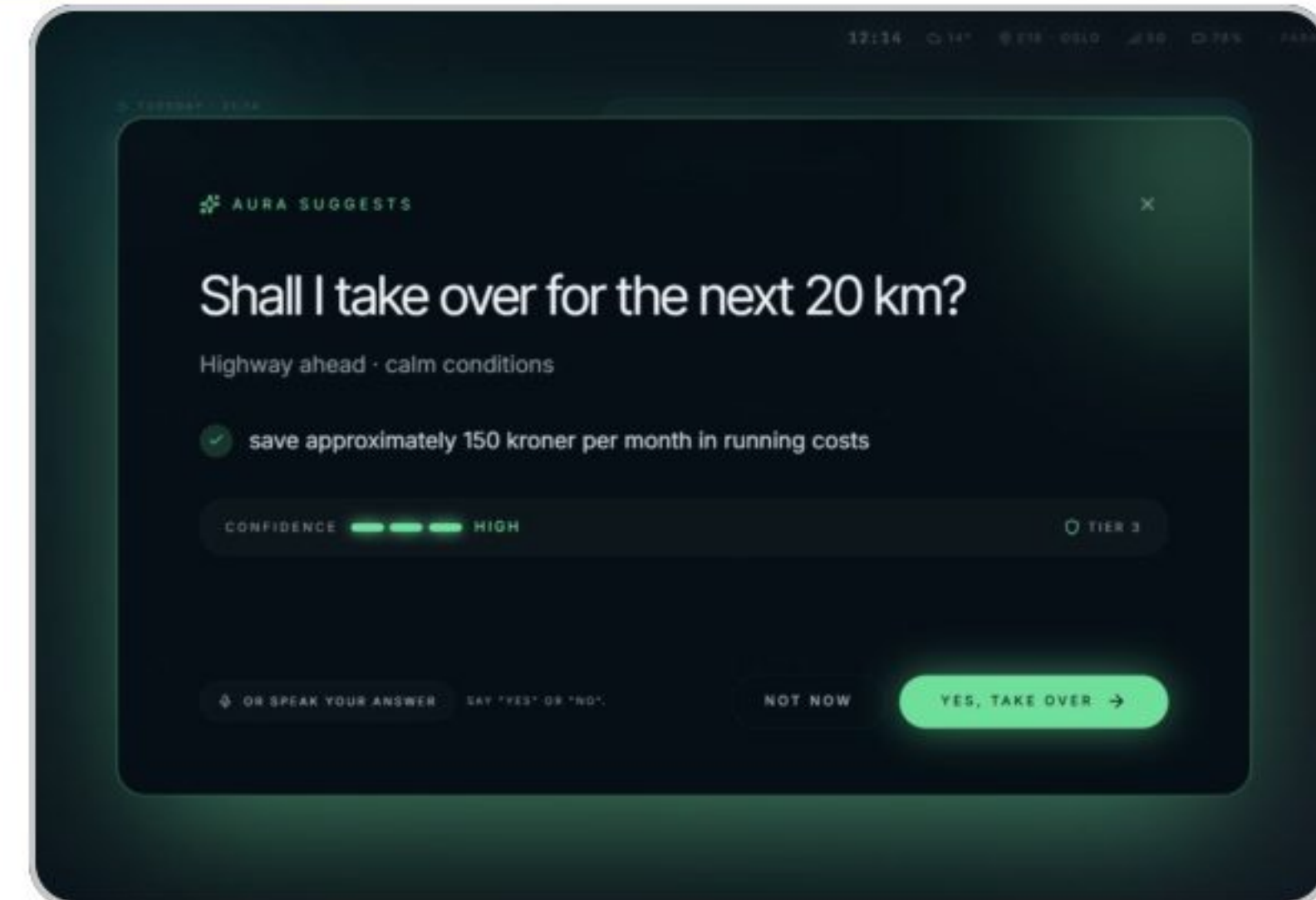
Building Trust

**01**

Driver Profile Setup (Onboarding)

It begins with short, optional questionnaire not to bore the driver. In the background the system monitors the driver to complete driver profile. Then minor questions also can be asked later to build up the profile.

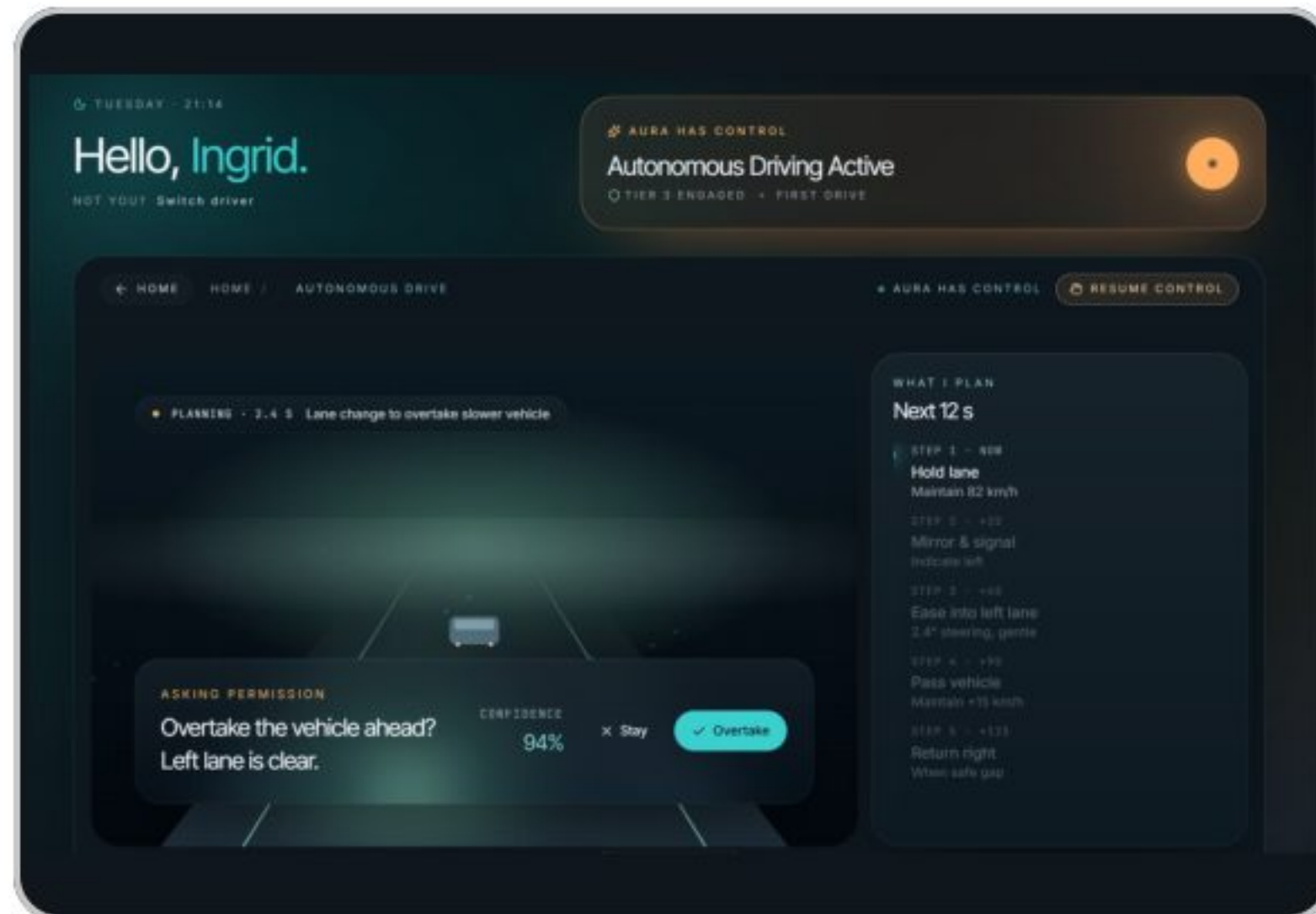
★ WOW FACTOR

**02**

Progressive Automation (Suggestions)

Based on the driver profile, gathered driving and behaviour data the system tries to adapt to driver and suggests driver to turn on autonomous driving and use higher levels of it.

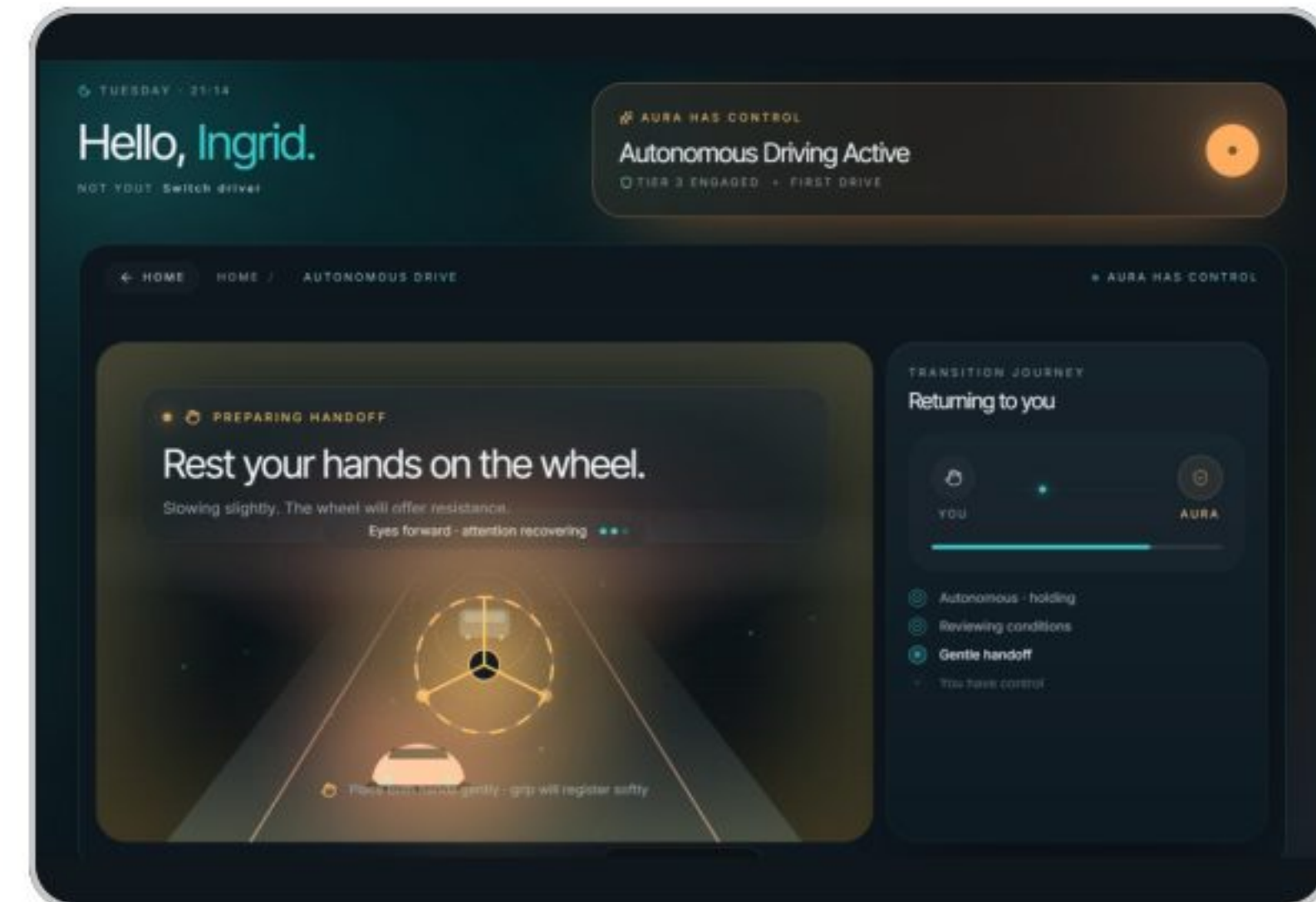
Building Trust



03

System Logic Presentation

When the autonomous driving is engaged the system shows the decisions it makes prior to execution (as far as possible) and possibly the reason behind them by a virtual reality of the road and the traffic around the car, graphical elements like path trajectory and minimal messages.



04

Optimal Takeover Sequence

When the autonomous driving system fails to continue operating (for sensor failure or unusual scenarios), the system tries to keep the driver calm yet attentive by adaptive and effective warning levels.

Interaction Flow

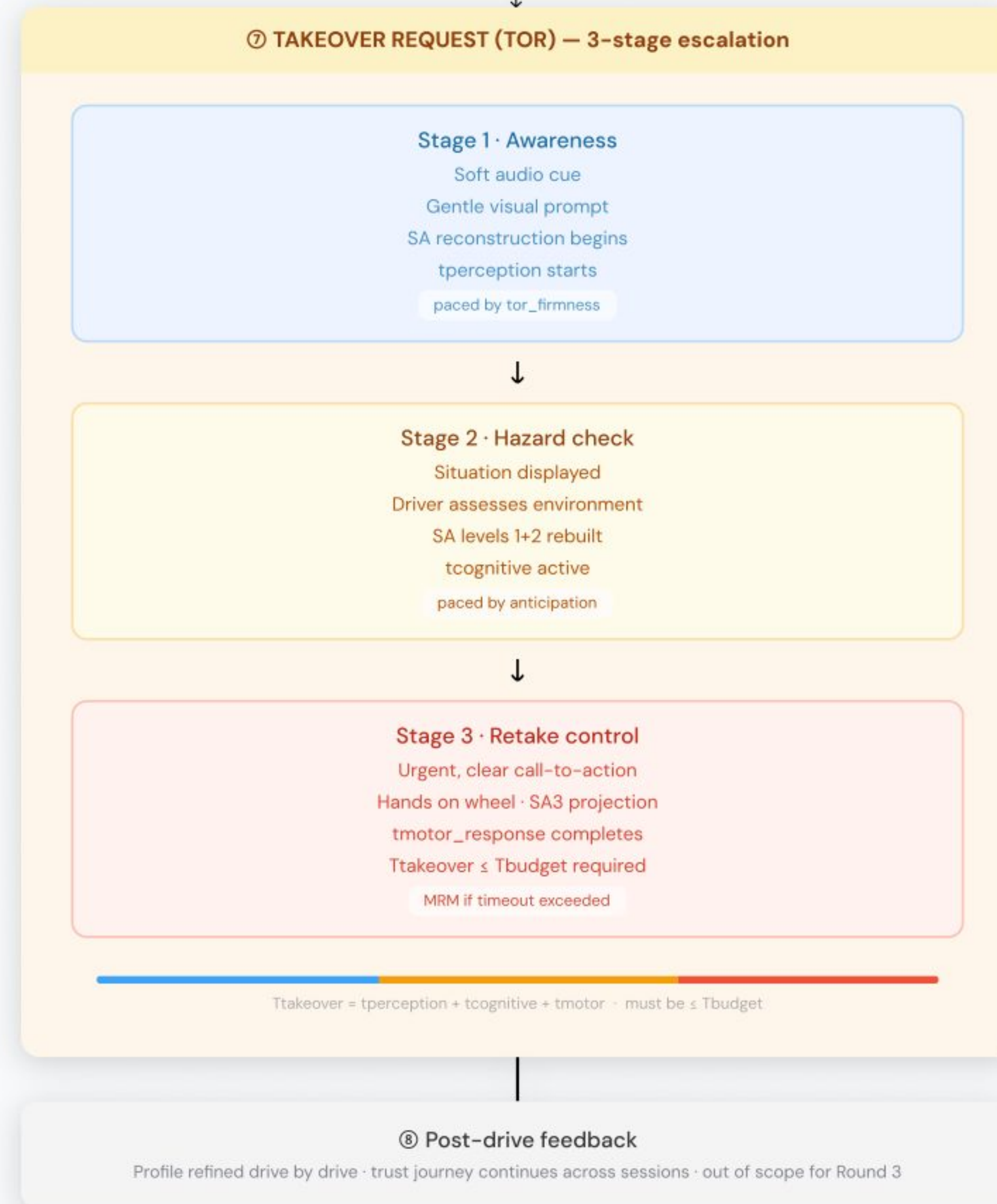
Interaction Flow

Building Trust



Interaction Flow

Flow



Implement

Features

Key Features

1. Safe handover under stress

When the system requests control back, I want clear progressive alerts, so I can take over without feeling startled.

2. Building trust progressively

When driving on a regular route, the system should recommend higher levels of autonomous driving and take the load off me over time

3. A system that adapts to me

When alerts don't match my style, I want the system to learn from my reactions, so I never reconfigure manually.

4. Understanding system intent

When the car makes an unexpected move, I want to see what it plans before it acts, so I feel informed and in control.

1. Safe handover under stress

WHEN

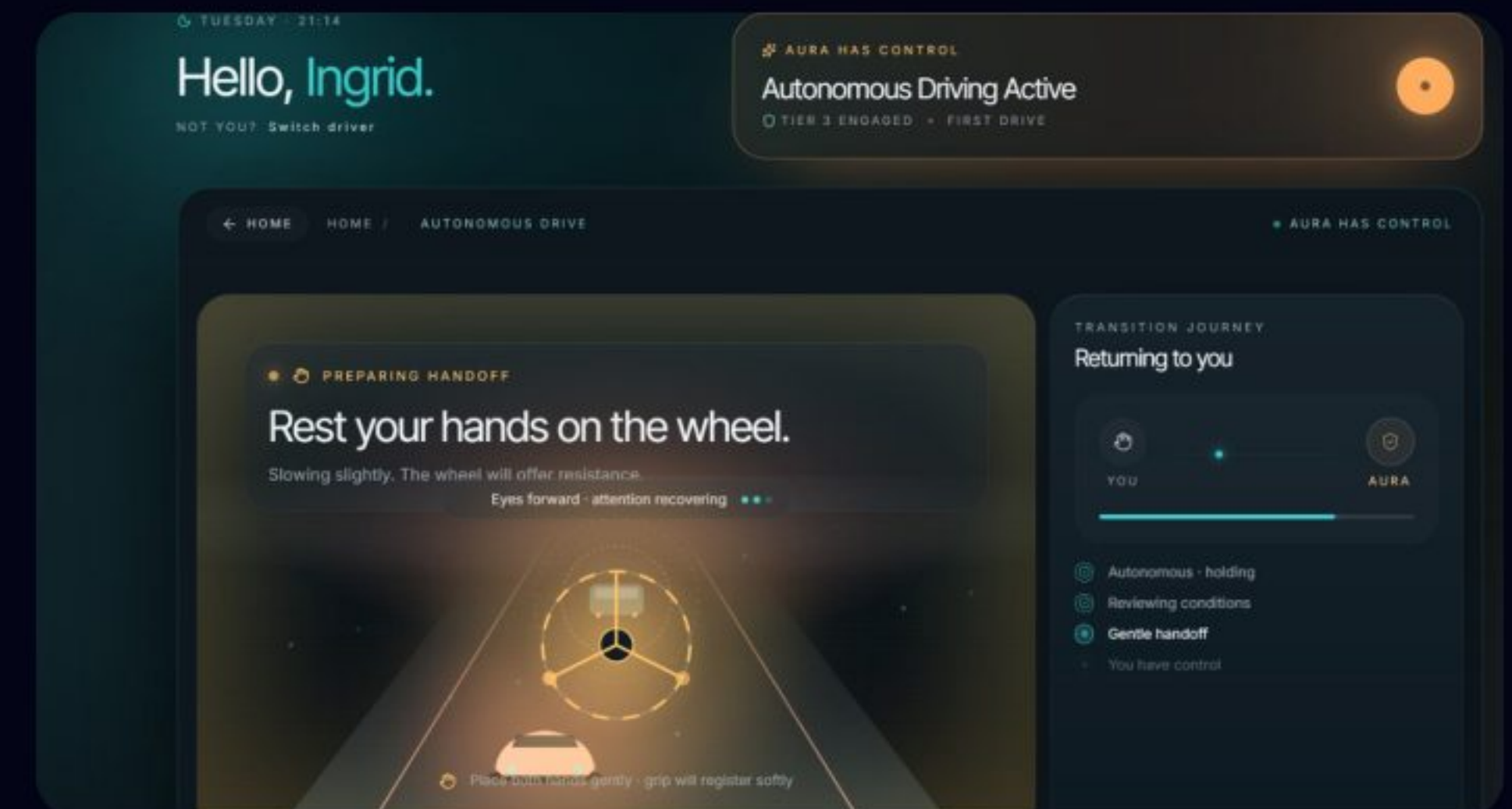
The autonomous system reaches its limits and requests the driver to take back control, while off-task, relaxed or distracted.

I WANT TO

Receive clear, progressive and non-startling alerts that give me enough time to re-engage before the situation becomes critical.

SO I CAN

Take over safely and calmly, without feeling startled or overwhelmed, even after being fully disengaged from driving.



2. Building trust progressively

WHEN

I am driving manually on a familiar highway route and conditions are ideal, light traffic, good visibility, long straight road.

I WANT TO

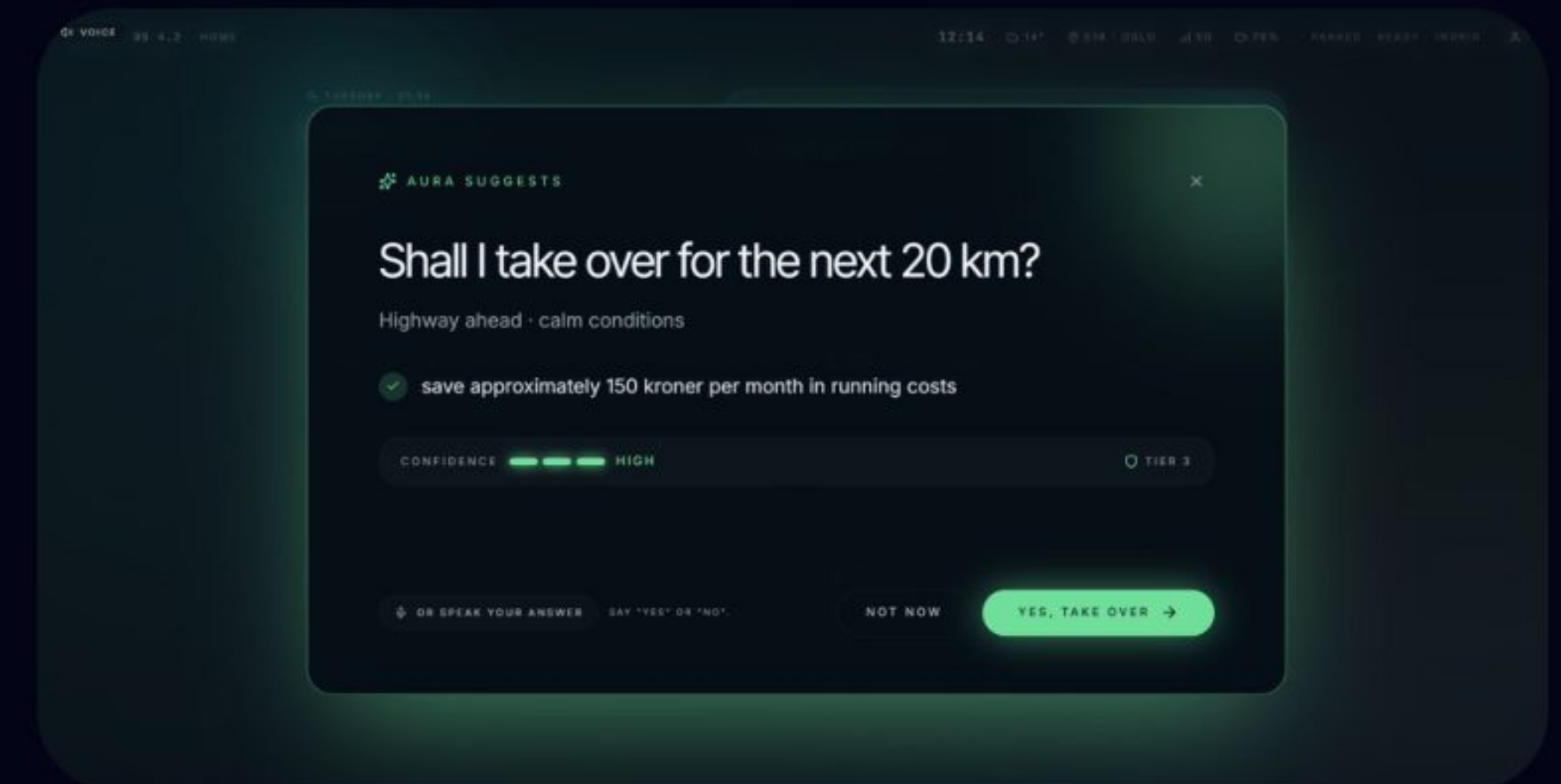
Receive a personalised contextual suggestion to switch to L3, with a concrete benefit, reduced fatigue, lower emissions, better travel time.

SO I CAN

Start trusting the system gradually, driven by a clear benefit, without having to decide on my own initiative when to hand over control.

Example suggestion on HMI

"Shall I take over for the next 20 km? This will save approximately 45 min of your time."



3. A system that adapts to me

WHEN

The system communicates in a way that does not match my comfort level, alerts too early, too intrusive, or not informative enough.

I WANT TO

Have the system learn from my reactions and adjust alert timing, modality and verbosity automatically, without manual settings.

SO I CAN

Experience an HMI that feels tailored to me from the first drive, improving naturally as the system understands my preferences.

Layer 1 — Onboarding

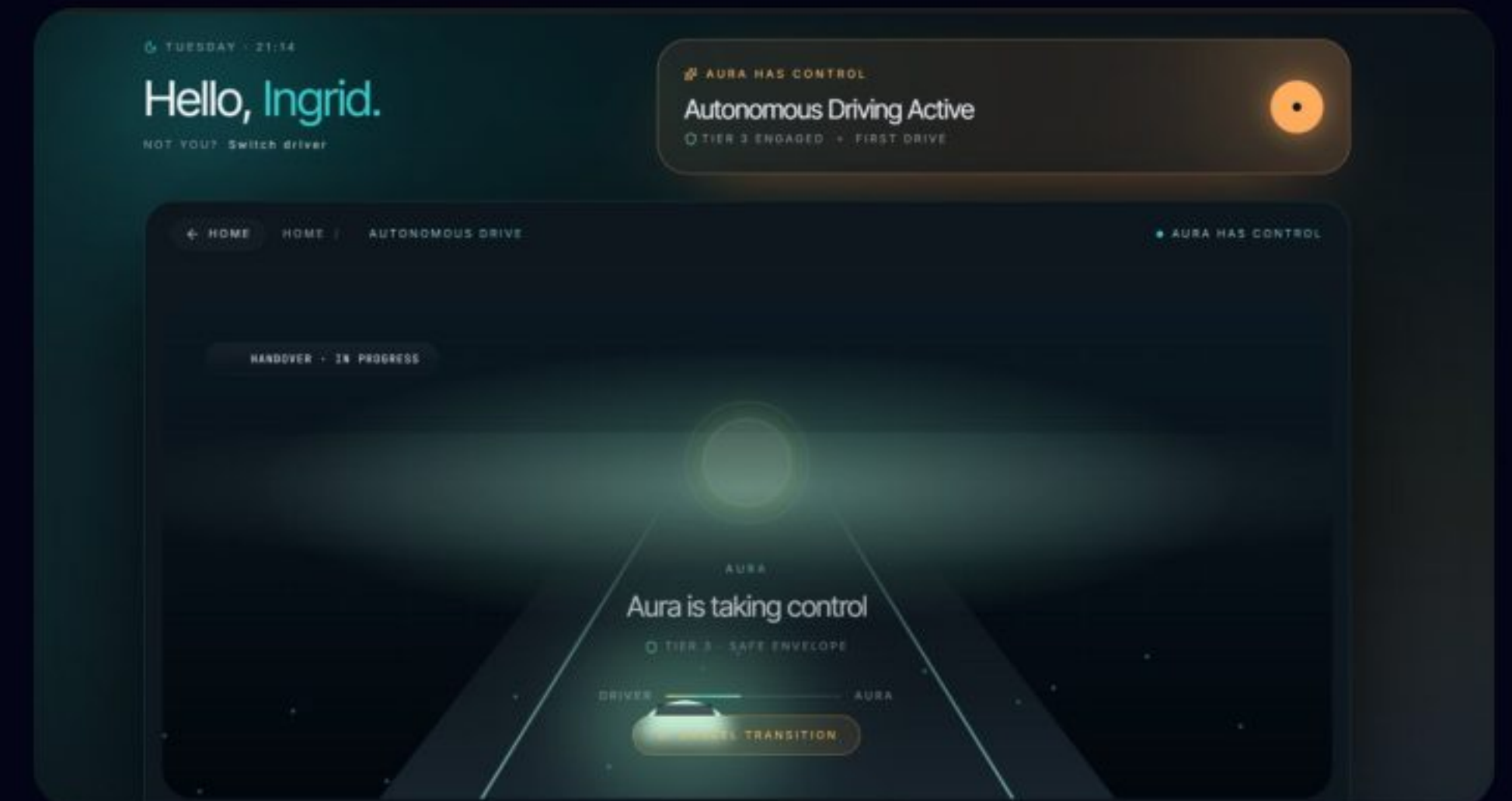
4 questions set the baseline profile before the first drive

Layer 2 — Background

Steering, speed and braking patterns refine the profile silently

Layer 3 — Refining

Response time and smoothness fine-tune alert timing after each event



4. Understanding system intent

WHEN

The car is driving autonomously and is about to make a decision I was not expecting, changing lanes, slowing down, or adjusting its route.

I WANT TO

See what the system is planning before it acts, a visual preview, a brief explanation, or an ambient cue that prepares me for the upcoming action.

SO I CAN

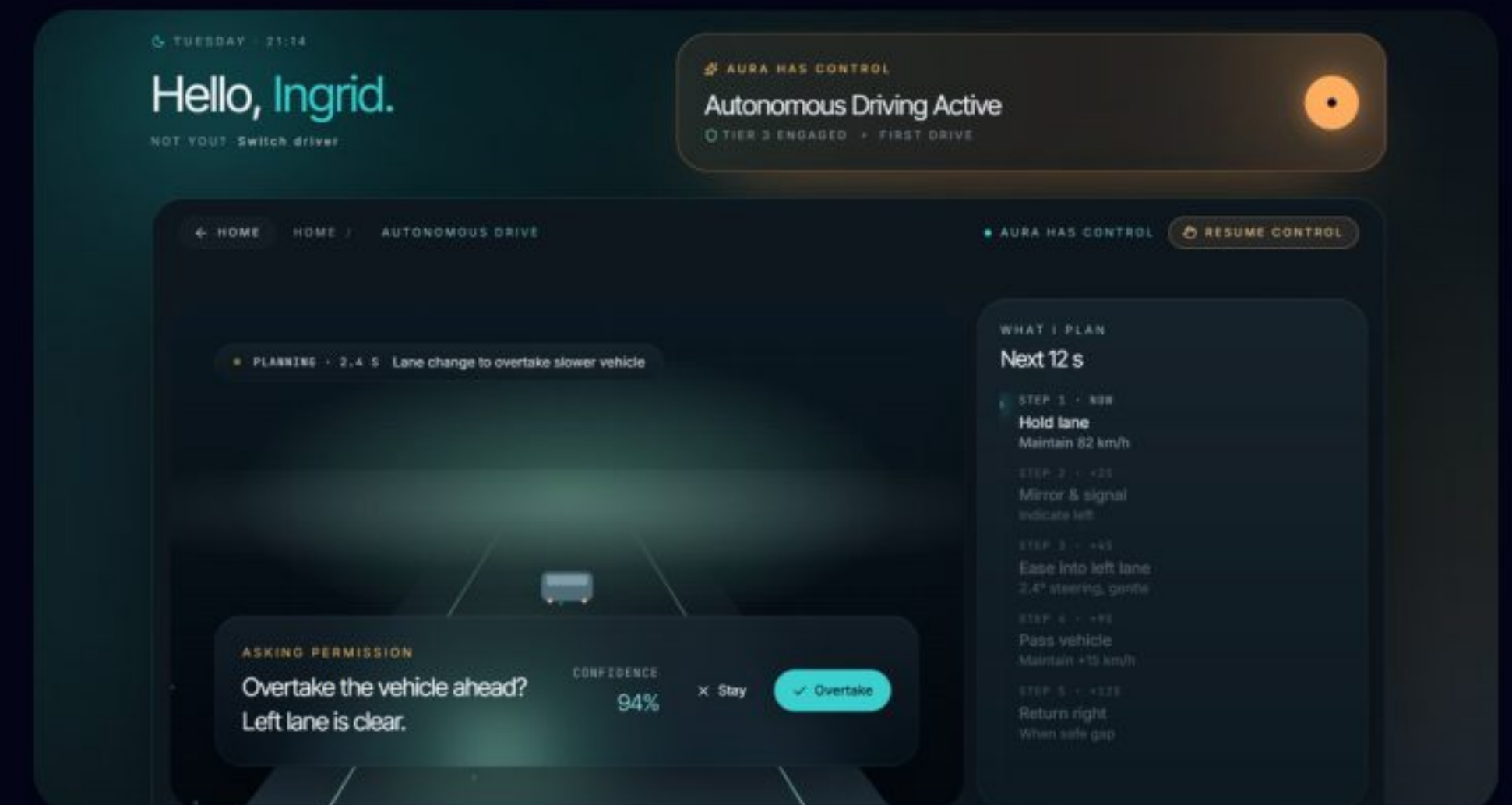
Feel informed and in control even when I am not driving, maintaining situational awareness and trust in the system's decisions.

Lane change

"Slower vehicle ahead, changing to the left lane in 3 seconds." Arc previews the full path before execution.

Speed reduction

"Speed limit changing ahead, reducing to 90 km/h." Confidence bar shifts and speed display updates smoothly.



Driver profile build

The Driver Profile

Driving Speed



Driving Style



Heads-up Timing



Takeover Pacing



Information Detail



How it updates

- Each spoken response is analysed by the LLM.
- Primary dimension shifts: $\pm X$ per interaction.
- Secondary (by inference): max $\pm X/2$.
- Endorsements never push the profile away.

—————> Five adaptive dimensions — each updated by every voice interaction. <—————

Four Phases, One Adaptive Profile

01 Onboarding

3 indirect lifestyle questions infer the initial driver profile. The driver never knows which dimension each question maps to — this reduces self-reporting bias.

02 Proactive L3 suggestion

Personalised popup when a highway segment is detected. Rotating benefits drawn from the driver profile: save time, reduce fatigue, lower fuel use.

03 In-drive voice commands

13 intents across music, climate, navigation and system queries. LLM extracts intent and parameters in real time from natural speech.

04 Post-drive feedback

3 post-drive questions shift the profile in the direction the driver actually prefers. Endorsements never push the profile away.



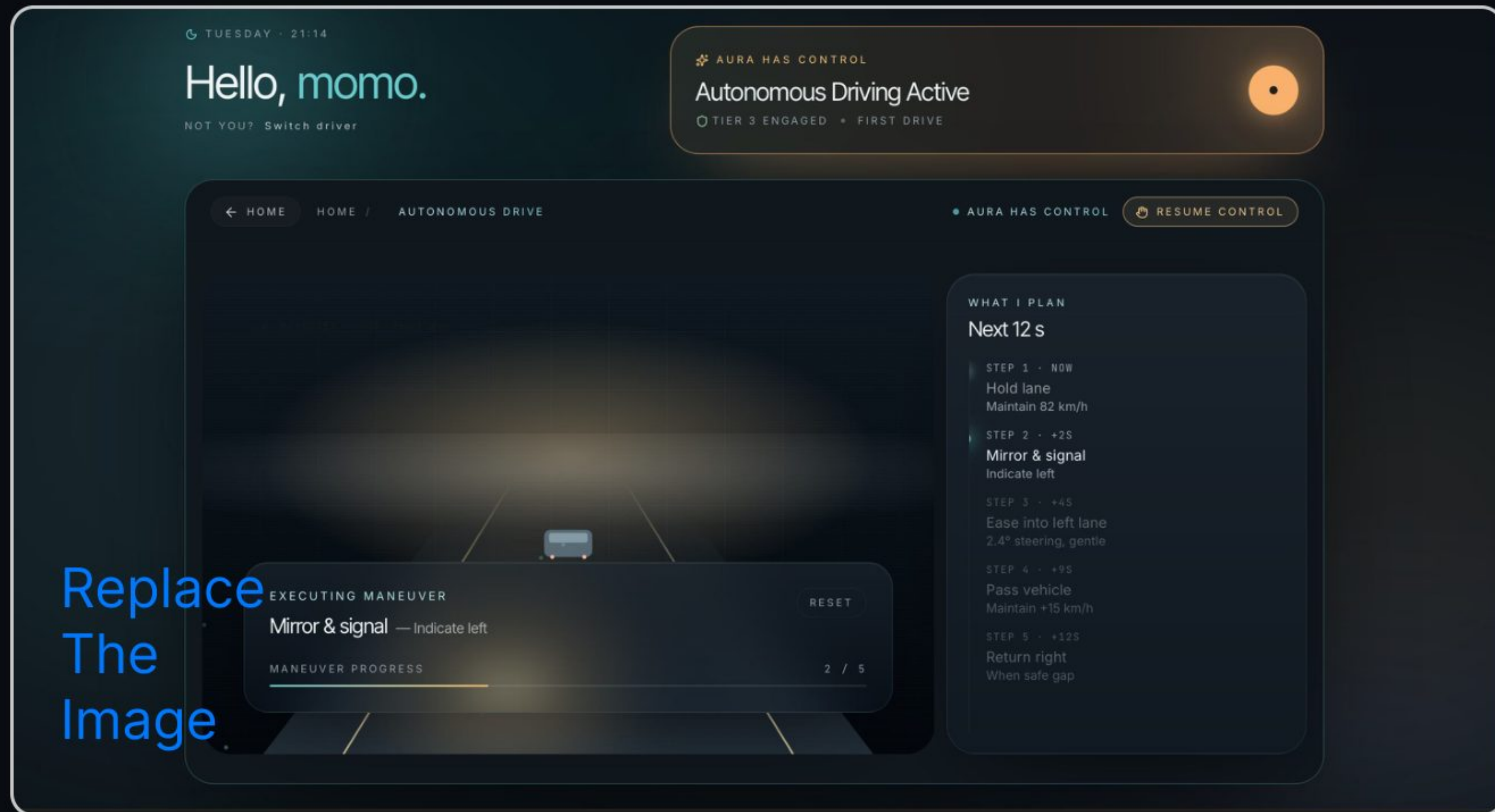
Prototype

Prototype

TRUSTDRIVE PROTOTYPE SETUP

GENERAL OVERVIEW





Prototype Link: <https://linklink> (replace this part with real link)

Testing Activity

Activity

How might we design Level 3 interactions that **build trust over time**, so drivers feel confident relying on the system later in life?

How We Test

- 01 Pre-test briefing**
Intro to L3 system
- 02 Onboarding**
5 indirect questions → driver profile built
- 03 Driving scenario**
Highway sim, suggestion popup, voice commands
- 04 Post-drive feedback**
Refine the adaptive profile
- 05 Debrief**
Open reflection on trust and comfort



Eye Tracking

Eye tracking revealed that the current HMI interface contains too much information, causing attention to scatter and posing a significant safety risk during takeover events.



Web prototype testing

In-class web prototype testing identified two key issues: the login flow lacked intelligence, prompting us to add fingerprint authentication; and the profile setup questions were revised to better capture driver preferences.

How We Measure

Success = HMW achieved when primary metric exceeds the threshold.

Methods: performance metrics · standardized self-report (Likert) · eye tracking — triangulated across quantitative sources

SUCCESS METRIC

Performance metric · Interaction log



Approved System Takeover

Formula

Accepted

÷

Total shown

× 100

Trust calibration

Standardized self-report · Likert

Self-reported trust in the system — before vs. after the session

Comfort rating

Standardized self-report · Likert

1–10 self-report after each system takeover

Eye tracking

Physiological · Gaze data

Fixation & dwell time on HMI elements during takeover events

Thanks.

Sources

- [1] Hansel, "Situation Awareness," SlideServe. Accessed: Jun. 13, 2026. [Online]. Available: <https://www.slideserve.com/Hansel/Situation-Awareness>
- [2] "AAA: Fear Of Self-Driving Cars On The Rise – AAA Newsroom." Accessed: Jun. 13, 2026. [Online]. Available: <https://newsroom.aaa.com/2023/03/aaa-fear-of-self-driving-cars-on-the-rise/>
- [3] "Driving Trust: Paving The Road For Autonomous Vehicles," World Economic Forum. Accessed: Jun. 13, 2026. [Online]. Available: <https://www.weforum.org/stories/2024/01/driving-trust-paving-the-road-for-autonomous-vehicles/>
- [4] F. Walker Et Al., "Trust In Automated Vehicles: Constructs, Psychological Processes, And Assessment," Front. Psychol., Vol. 14, Nov. 2023, Doi: 10.3389/fpsyg.2023.1279271.
- [5] A. Kuznietsov, B. Gyevar, C. Wang, S. Peters, And S. V. Albrecht, "Explainable AI For Safe And Trustworthy Autonomous Driving: A Systematic Review," IEEE Trans. Intell. Transport. Syst., Vol. 25, No. 12, Pp. 19342–19364, Dec. 2024, Doi: 10.1109/TITS.2024.3474469.
- [6] D. Omeiza, H. Webb, M. Jirotko, And L. Kunze, "Explanations In Autonomous Driving: A Survey," IEEE Transactions On Intelligent Transportation Systems, Vol. 23, No. 8, Pp. 10142–10162, Aug. 2022, Doi: 10.1109/TITS.2021.3122865.